Appendix

Mechanics

In the agent-based simulations, a single cell i is represented by the position of its centre of mass x_i and its dynamics is given by the following equation of motion [1, 2]:

$$m_i \frac{d\mathbf{v}_i}{dt} = -\lambda_s \mathbf{v}_i - \lambda_r \sum_j (\mathbf{v}_i - \mathbf{v}_j) + \sum_j (\mathbf{F}^s_{ij} + \mathbf{F}^a_{ij}) \tag{1}$$

where isotropic friction forces are assumed and $\mathbf{v}_i$ is the cell velocity. λ_s is the friction coefficient for the cell-substrate or cell-ECM interaction, while λ_r is the relative cell-cell friction coefficient.

$\mathbf{F}^s_{ij}(\mu, \nu)$ is a passive force that describes cell-cell adhesion and repulsion that depends on the position of the two interacting cells (i, j) as well as their states (μ, ν):

$$\mathbf{F}^s_{ij}(\mu,\nu) = \begin{cases} F^{(\mu,\nu)}_{rep}(\frac{2r}{d_{ij}} - 1)(\frac{\mu 2r}{d_{ij}} - 1)\frac{(\mathbf{x}_i - \mathbf{x}_j)}{d_{ij}} & if\ d_{ij} < 2r \\ F^{(\mu,\nu)}_{adh}(\frac{2r}{d_{ij}} - 1)(\frac{\mu 2r}{d_{ij}} - 1)\frac{(\mathbf{x}_i - \mathbf{x}_j)}{d_{ij}} & if\ 2r < d_{ij} < 2\mu r \\ 0 & otherwise \end{cases} \tag{2}$$

where $F^{(\mu,\nu)}_{rep}$, $F^{(\mu,\nu)}_{adh}$ are 3×3 matrices that describe the magnitude of the interaction and d_{ij}, $(\mathbf{x}_i - \mathbf{x}_j)$ are the distance between cells i and j and the distance vector. r is a parameter used to set the characteristic length-scale of the simulations while the parameter μ is used to set the adhesion force interaction range.

$\mathbf{F}^a_{ij}(\mu, \nu)$ is the cell-cell active protrusion force, corresponding to a dipolar contractile interaction between two cells:

$$\mathbf{F}^a_{ij}(\mu,\nu) = \begin{cases} F^{(\mu,\nu)}_P \frac{(\mathbf{x}_i - \mathbf{x}_j)}{d_{ij}} & if\ 2r < d_{ij} < \mu_P r \\ 0 & otherwise \end{cases} \tag{3}$$

where $F_P^{(\mu,\nu)}$ is the Similarly to μ, μ_P is a parameter used to set the protrusion force interaction range. However, in order for a protrusion bond to be *established* between two cells, the following condition must be met:

$$2\mu r < d_{ij} < \mu_P r \tag{4}$$

This is done to provide an accurate representation of aggregate fluidity by promoting the exchange of cells that do not necessarily interact via the adhesion force.

We set a protrusion force activation rate $k_{p_{\text{on}}}$ and deactivation rate $k_{p_{\text{off}}}$, which are specific to each cell fate, whereby only B-type and C-type cells have $k_{p_{\text{on}}} \neq 0$ and are therefore capable of attempting to form a bond with another cell. B-type cells interact through protrusion forces exclusively with other B-type cells, while A-type and C-type cells can form protrusion interactions with each other. As a first approximation, each cell is allowed to actively extend a single protrusion at any given time, but there is no restriction on the number of protrusions it may receive from neighbouring cells. Consequently, a cell can participate in multiple protrusive interactions while actively sending out only one protrusion itself.

The probability of establishing a bond between a pair of cells is therefore given by: $P = k_{p_{\text{on}}} \cdot \Delta t$, where Δt is the time-step used in the simulations; and the duration of the contractile interaction is modelled as:

$$t_{ij} = -\frac{log(\xi)}{k_{p_{\text{off}}}} \tag{5}$$

where ξ is a random number drawn from a uniform distribution $\text{U}(0, 1)$.

Cell division

Cells undergo division according to the following rules:

- A division axis is randomly chosen over the unit sphere.
- The two daughter cells are generated with equal radius as the parent cell and positioned on opposite sides of the division axis which is centred at the parent cell's location. The daughter cells are generated with their centre of mass at a distance of 40% from that of the parent cell.
- A new division time is then assigned to each daughter cell, sampled from a uniform distribution with mean τ_{div} and standard deviation σ_{div}:

$$\mathrm{U}(\tau_{div}(1-\sigma_{div}), \tau_{div}(1+\sigma_{div})) \tag{6}$$

To prevent artificially large velocities caused by strong repulsive forces immediately after cell division, each daughter cell is assigned a relaxation time τ_{rel}. During this period, the daughters do not contribute to their neighbours' relative friction in Eq. (1). The relaxation time roughly corresponds to the number of time steps until the daughter cell has accumulated N_{rel} neighbour counts, where the same neighbour cell can be counted at successive steps For example, a cell with only one neighbour for 50 consecutive steps will have $\tau_{rel} = 50\Delta t$, whereas a cell surrounded by more than 50 neighbours completes its relaxation time within a single step.

when the cumulative number of interactions reaches N_{rel}

Cell differentiation

Cell differentiation in the simulations follows the 3-state differentiation model explained in the main text:

Derivation of the equation of motion for the simulations

Cells find themselves in the overdamped, low Reynolds number regime [3]. Here, viscous forces dominate over inertial forces and therefore: $m_i \frac{dv_i}{dt} \to 0$. This allows us to rewrite equation (1) as:

$$\lambda_s \mathbf{v}_i + \lambda_r \sum_j (\mathbf{v}_i - \mathbf{v}_j) = \sum_j (\mathbf{F}^s_{ij} + \mathbf{F}^a_{ij}) \tag{7}$$

$$\frac{\lambda_s}{\lambda_r}\mathbf{v}_i + N_j \mathbf{v}_i - \sum_j \mathbf{v}_j = \frac{1}{\lambda_r} \sum_j (\mathbf{F}^s_{ij} + \mathbf{F}^a_{ij}) \tag{8}$$

$$\Big(\frac{\lambda_s}{\lambda_r N_j} + 1\Big)\mathbf{v}_i = \frac{1}{N_j \lambda_r} \sum_j (\mathbf{F}^s_{ij} + \mathbf{F}^a_{ij}) + \frac{1}{N_j} \sum_j \mathbf{v}_j \tag{9}$$

As our simulations focus on 3D suspended cell aggregates, we assume $\lambda_s \approx 0$. Often, the λ_s term is omitted entirely [4, 5],

however, retaining a small but non-zero value is essential for numerical stability. Accordingly, we take $\lambda_s \ll \lambda_r$ with $\lambda_s \neq 0$, which allows us to define:

$$\epsilon \equiv \frac{\lambda_s}{\lambda_r N_j}, \tag{10}$$

so that ϵ can be arbitrarily small. We then introduce:

$$\alpha \equiv \frac{1}{\epsilon + 1} \lesssim 1, \quad \text{and} \quad \lambda \equiv \frac{\alpha}{\lambda_r}. \tag{11}$$

With these definitions, the velocity of cell i can be written as

$$\mathbf{v}_i = \frac{\lambda}{N_j} \sum_j \left(\mathbf{F}^s_{ij} + \mathbf{F}^a_{ij}\right) + \frac{1}{N_j} \sum_j \mathbf{v}_j \alpha. \tag{12}$$

where λ is the effective force coefficient and α can be thought of as a velocity dissipation term for the neighbour cells.

Non-dimensionalisation

By non-dimensionalising equation (12) with respect to a typical distance R_0, force F_0 and timescale T_0, we obtain the equation of motion that is used directly for the simulations:

$$\tilde{\mathbf{v}}_i = \frac{\tilde{\lambda}}{N_j} \sum_j (\tilde{\mathbf{F}}^s_{ij} + \tilde{\mathbf{F}}^a_{ij}) + \frac{1}{N_j} \sum_j \tilde{\mathbf{v}}_j \alpha \tag{13}$$

where $\tilde{\lambda} \equiv \lambda R_0/(F_0 T_0)$, $\tilde{v}_i \equiv v_i T_0/R_0$ and $\tilde{F}_{ij} \equiv F_{ij}/F_0$. A similar approach is carried out in order to non-dimensionalise all the parameters in the simulations, which are denoted by the $\sim$ symbol.

$R_0 = 5$ μm is taken as the typical cell radius in gastruloids (Gros et al. Biorxiv), $T_0 = R_0/V_0 \simeq 10$ min , where $V_0 \simeq 0.6$ μm/min is the typical cell instantaneous speed in gastruloids(Anlas et al Biorxiv 2025), and $F_0 = 100$ pN is the typical cell adhesion force.

Additional figures concerning cell packing

To study the cell packing and nearest-neighbour count per cell, we use the 3D Delaunay triangulation method [6] with a cutoff distance of $2\tilde{r}$ (where $\tilde{r}$ is the characteristic length of the simulations) in order to consider only cells that interact via volume exclusion as neighbours. Fig. S1 shows a typical example from simulations of how the average cell-cell distance $\langle \tilde{d}_{ij} \rangle$ evolves with time for each cell fate and for the whole aggregate, whereas Fig. S2 gives the time evolution of the average number of neighbours per cell $\langle N \rangle$.

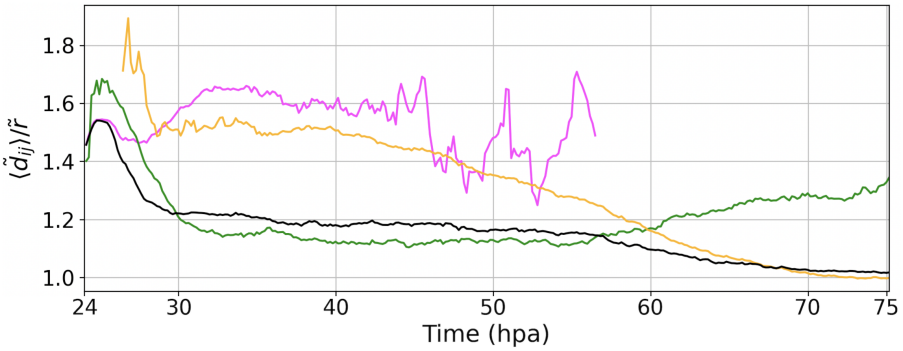


Figure S1: Time evolution of the average cell-cell distance $\langle \tilde{d}_{ij} \rangle$ throughout the simulation using Delaunay triangulation with a cutoff distance of $2\tilde{r}$ for: cell fate A (magenta); B (green); C (yellow); and the whole aggregate (black).

Fig. S1 provides an intuitive definition of an average effective cell radius, $\langle \tilde{R} \rangle$ by taking: $\langle \tilde{R} \rangle \sim \langle \tilde{d}_{ij} \rangle / 2$. The drop in $\langle \tilde{d}_{ij} \rangle$ from around 25hpa to 30hpa corresponds to the core formation of the B-type (T+) cells. At around 40hpa the polarisation process has begun and the T+ cluster begins to move towards the periphery of the gastruloid. During this process, $\langle \tilde{d}_{ij} \rangle$ of the T+ cluster fluctuates around a constant value, reflecting a constant density. In the wake of the T+ cluster C-type cells can now cluster together themselves, decreasing their $\langle \tilde{d}_{ij} \rangle$ and increasing their packing. (In the current state of the model, the density of a homogeneous cluster of cells grows with the number of cells in the cluster. This happens because as the cluster grows due to cell proliferation, there will be higher mechanical pressure, which mostly affects the innermost cells of the cluster moving radially inwards towards its core). At around 60hpa, the T+ cluster has reached the periphery and the elongation process begins. As the cluster exits the C-type cluster $\langle \tilde{d}_{ij} \rangle$ begins to increase, marking a decrease in density of the T+ cluster. Finally, at around 70hpa, the elongation process ceases and the $\langle \tilde{d}_{ij} \rangle$ values for each cell type reach a plateau indicating stable densities for both the B-type and C-type regions in the gastruloid before the T+ cluster begins to progressively have less cell as they differentiate to C-type cells. (Note that the increase in the fluctuations of the A-type cell population $\langle \tilde{d}_{ij} \rangle$ from around 45hpa onwards is due to having a very small number of them).

The density differences in each cell population can also be appreciated by studying Fig. S2, where a higher density (smaller $\langle \tilde{d}_{ij} \rangle$ value) is reflected by a higher average number of nearest-neighbours $\langle N \rangle$.

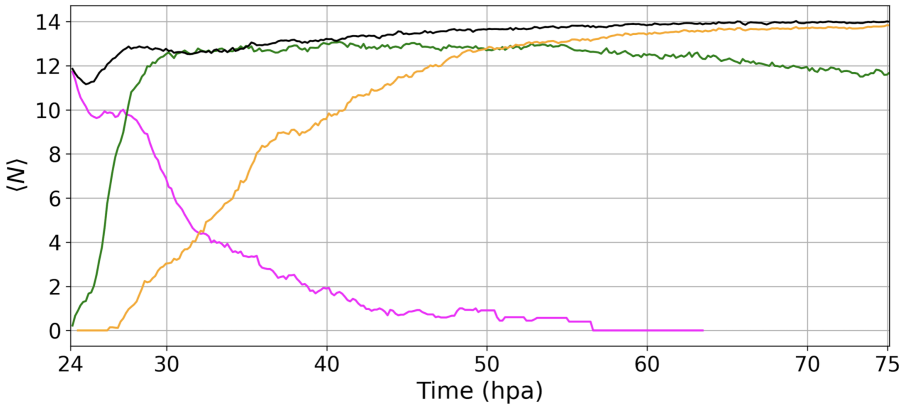


Figure S2: Time evolution of the average number of neighbours per cell $\langle N \rangle$ throughout the simulation using Delaunay triangulation with a cutoff distance of $2\tilde{r}$ for: cell fate A (magenta); B (green); C (yellow); and the whole aggregate (black).

Simulation parameter values

Table S1: Scalar parameters.

Parameter	Symbol	Value
Time step	$\tilde{\Delta}t$	0.01
Characteristic length	$\tilde{r}$	1
Passive force range paramter	μ	2
Protrusion force range paramter	μ_P	$4\sqrt{3}$ (~ 7)
Average cell division time	$\tilde{\tau}_{\text{div}}$	120
SD of cell division time	$\tilde{\sigma}_{\text{div}}$	0.5
Effective force coefficient	$\tilde{\lambda}$	1
Neighbour velocity dissipation	α	0.98
$A \to B$ transition rate	$\tilde{p}$	0.03
$B \to C$ transition rate	$\tilde{q}$	0.01
Cell division relaxation time	N_{rel}	50
Cell division inhibition factor	K	0.2

Table S2: Adhesion pair-interaction coefficients $\tilde{F}_{adh}$ between different cell types.

	A	B	C
A	0.3	0.05	0.3
B	0.05	1.5	0.05
C	0.3	0.05	0.3

Table S3: Repulsion pair-interaction coefficients $\tilde{F}_{rep}$ between cell types.

	A	B	C
A	0.5	0.5	0.5
B	0.5	1.5	0.5
C	0.5	0.5	0.5

Table S4: Protrusion force pair-interaction coefficients $\tilde{F}_P$ between cell types.

	A	B	C
A	5	0	5
B	0	5	0
C	5	0	5

Table S5: Protrusion force activation rate $k_{p_{\text{on}}}$.

A	B	C
0	2	2

Table S6: Protrusion force deactivation rate $k_{p_{\text{off}}}$.

	A	B	C
A	2	20	2
B	20	0.4	20
C	2	20	2

References

[1] Paul Van Liedekerke, MM Palm, Nick Jagiella, and Dirk Drasdo. Simulating tissue mechanics with agent-based models: concepts, perspectives and some novel results. *Computational particle mechanics*, 2(4):401–444, 2015.

[2] Brian A Camley and Wouter-Jan Rappel. Physical models of collective cell motility: from cell to tissue. *Journal of physics D: Applied physics*, 50(11):113002, 2017.

[3] Edward M Purcell. Life at low reynolds number. In *Physics and our world: reissue of the proceedings of a symposium in honor of Victor F Weisskopf*, pages 47–67. World Scientific, 2014.

[4] Paul Van Liedekerke, Johannes Neitsch, Tim Johann, Kévin Alessandri, Pierre Nassoy, and Dirk Drasdo. Quantitative cell-based model predicts mechanical stress response of growing tumor spheroids over various growth conditions and cell lines. *PLoS computational biology*, 15(3):e1006273, 2019.

[5] Paul Van Liedekerke, Johannes Neitsch, Tim Johann, Enrico Warmt, Ismael Gonzàlez-Valverde, Stefan Hoehme, Steffen Grosser, Josef Kaes, and Dirk Drasdo. A quantitative high-resolution computational mechanics cell model for growing and regenerating tissues. *Biomechanics and modeling in mechanobiology*, 19(1):189–220, 2020.

[6] Franco P Preparata and Michael I Shamos. *Computational geometry: an introduction*. Springer Science & Business Media, 2012.